

Exercise - 7 Solutions

Calibration Concepts

Exercise 7.1: Epistemic vs. Aleatoric Uncertainty

Define epistemic and aleatoric uncertainty.

Type	Meaning	Can it be reduced?	Example
Epistemic uncertainty	Uncertainty caused by lack of knowledge or lack of training data	Yes, it can be reduced with more data, better features, or better model coverage	A model trained only on daytime images sees a night image
Aleatoric uncertainty	Uncertainty caused by inherent noise or ambiguity in the data	No, it remains even with more data	A blurry image, sensor noise, ambiguous labels

1. Which type is more relevant to OOD inputs (e.g., night images for a model trained on day images)?

→ Epistemic uncertainty is more relevant as the model trained only on day images, night images will be OOD for the model

2. Which type would you expect to be dominant in a correctly classified, in-distribution Image?

→ Although, uncertainty should usually be low as the image has already been correctly classified. But if some uncertainty remains, it is more likely to be aleatoric uncertainty.

The image comes from the same distribution as the training data, so the model should already have knowledge about it. Any remaining uncertainty is more likely due to unavoidable ambiguity, such as blur, partial occlusion, sensor noise, or unclear labels.

Exercise 7.2: Calibration and ECE

Explain what it means for a classifier to be well-calibrated. Define the Expected Calibration Error (ECE) and describe how it is computed.

→ A classifier is well-calibrated when its predicted confidence matches its actual correctness.

Calibration is the alignment between the model's confidence and its empirical accuracy. For example, if a weather model predicts "70% rain" many times, it should actually rain around 70% of those times.

ECE → Expected Calibration Error, is a number that measures how far the model is from perfect calibration.

Defined as - average gap between confidence and accuracy across bins, weighted by the number of samples in each bin.

$$ECE = \sum_{m=1}^M \frac{|B_m|}{n} |acc(B_m) - conf(B_m)|$$

Symbol	Meaning
(M)	Number of confidence bins
(B _m)	Set of predictions in bin (m)
(n)	Total number of samples
(acc(B _m))	Accuracy of predictions in bin (m)
(conf(B _m))	Average confidence of predictions in bin (m)

Bins with more samples contribute more strongly.

A lower ECE means better calibration, while a higher ECE means the model's confidence does not match its actual accuracy.

Exercise 7.3: Cost-Optimal Downstream Decisions

A pedestrian detector outputs $p = p(\text{ped} \mid x) \in [0, 1]$. The autopilot must choose between two actions with asymmetric costs:

Action	Pedestrian present	No pedestrian
Brake	0	$(C_{\text{FP}}=1)$
Continue	$(C_{\text{FN}}=100)$	0

At which p should the autopilot brake in order to minimize the costs?

1. Write down the expected loss (costs) of each action as a function of p : $E[L \mid \text{brake}]$ and $E[L \mid \text{continue}]$.

→

Consider,

p	Probability that pedestrian is present
$1 - p$	Probability that no pedestrian is present

Loss if the car brakes, braking only costs something when there is no pedestrian.

$$E[L \mid \text{brake}] = p \cdot 0 + (1-p) \cdot 1$$

$$E[L \mid \text{brake}] = 1 - p$$

Loss if the car continues, continuing only costs something when there actually is a pedestrian.

$$E[L \mid \text{continue}] = p \cdot 100 + (1-p) \cdot 0$$

$$E[L \mid \text{continue}] = 100p$$

2. Find the threshold τ^* at which both actions have equal expected loss. For $p > \tau^*$ the autopilot should brake; for $p < \tau^*$ it should continue.

→

$$E[L \mid \text{brake}] = E[L \mid \text{continue}]$$

$$1 - p = 100p$$

$$p = 1/101 = 0.0099$$

Consider - $\tau^* = 0.0099$

This threshold is the point where the decision changes.

If: $p > \tau^*$, then braking is cheaper.

If: $p < \tau^*$, then continuing is cheaper.

3. Compute τ^* for the costs above. How does it compare to the standard argmax rule ($\tau = 0.5$)?

→

$$\tau^* = \frac{C_{FP}}{C_{FP} + C_{FN}}$$

$$\tau^* = 1/101 = 0.0099$$

→ This means the car should brake even if the model gives only about a 1% pedestrian probability.

This threshold is much lower than 0.5 because missing a pedestrian is much more dangerous than braking unnecessarily. However, this only works if the model is well-calibrated; otherwise, the probability p may not reflect the true risk.

The normal 0.5 threshold treats false positives and false negatives as equally important. But here, a false negative is 100 times worse than a false positive. Therefore, the optimal threshold becomes much lower than 0.5.

→ The autopilot should brake when: $p > 0.0099$

So, if the pedestrian detector gives even about a 1% probability that a pedestrian is present, the autopilot should brake.

4. Why does τ^* only yield the optimal decision when p is well-calibrated? What can go wrong when the same threshold is applied to an overconfident model?

→ The threshold τ^* is optimal only when the predicted probability p is well-calibrated. This means that if the model says there is a 1% chance of a pedestrian, then the real frequency should also be about 1%.

If the model is overconfident, it may output very low pedestrian probabilities even when pedestrians are present. Then the autopilot may continue driving because $p < \tau^*$, although braking would have been the safer action. Therefore, a cost-optimal threshold only works when the model probabilities are reliable.

An overconfident model may produce very low probability for pedestrians even when a pedestrian is actually present.

The cost-optimal threshold only works if the model's predicted probability actually reflects real-world likelihood.

Exercise 7.4: Measuring Calibration

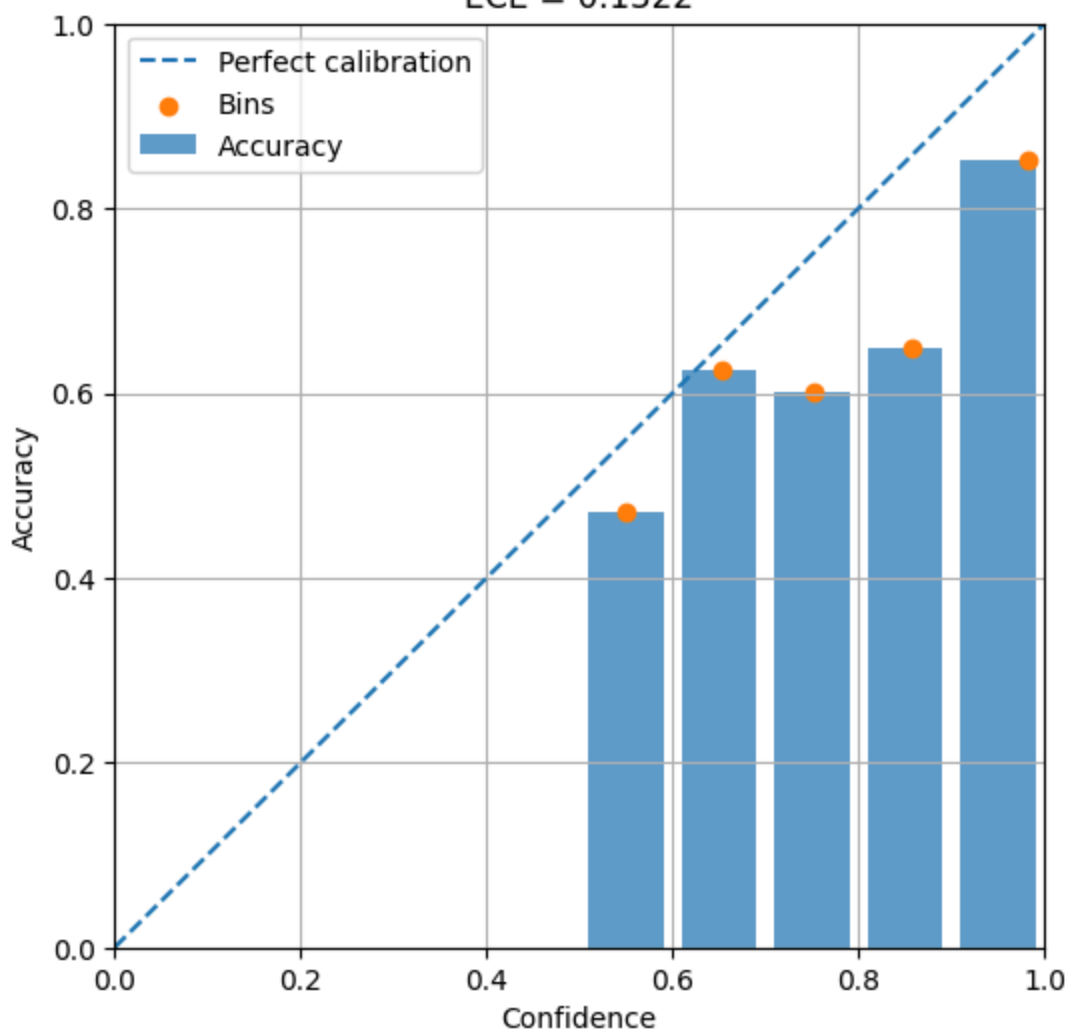
1. Compute the ECE of each of your three trained CARLA models on the in-distribution test set.

→ (refer ipynb for implementation)

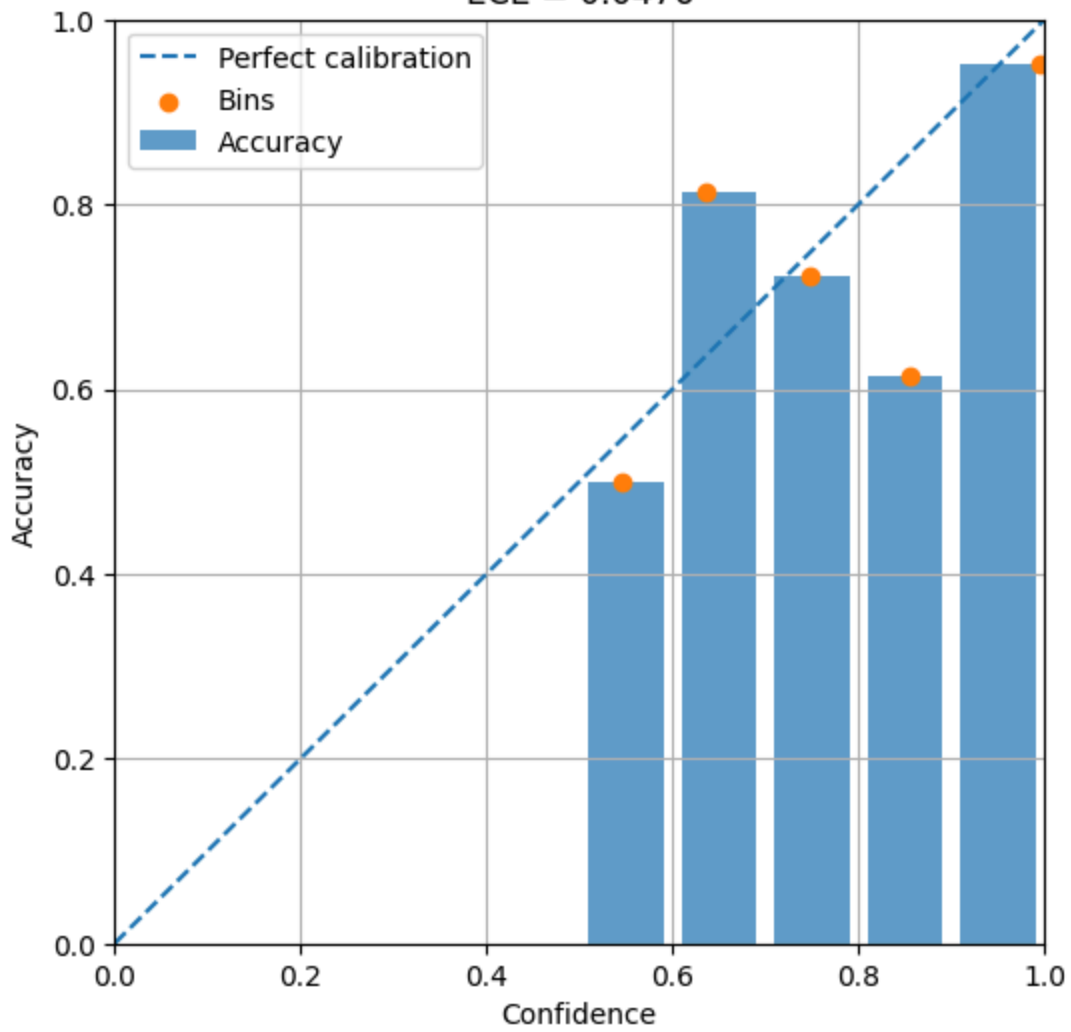
2. Plot a reliability diagram (confidence vs. accuracy) for each model.

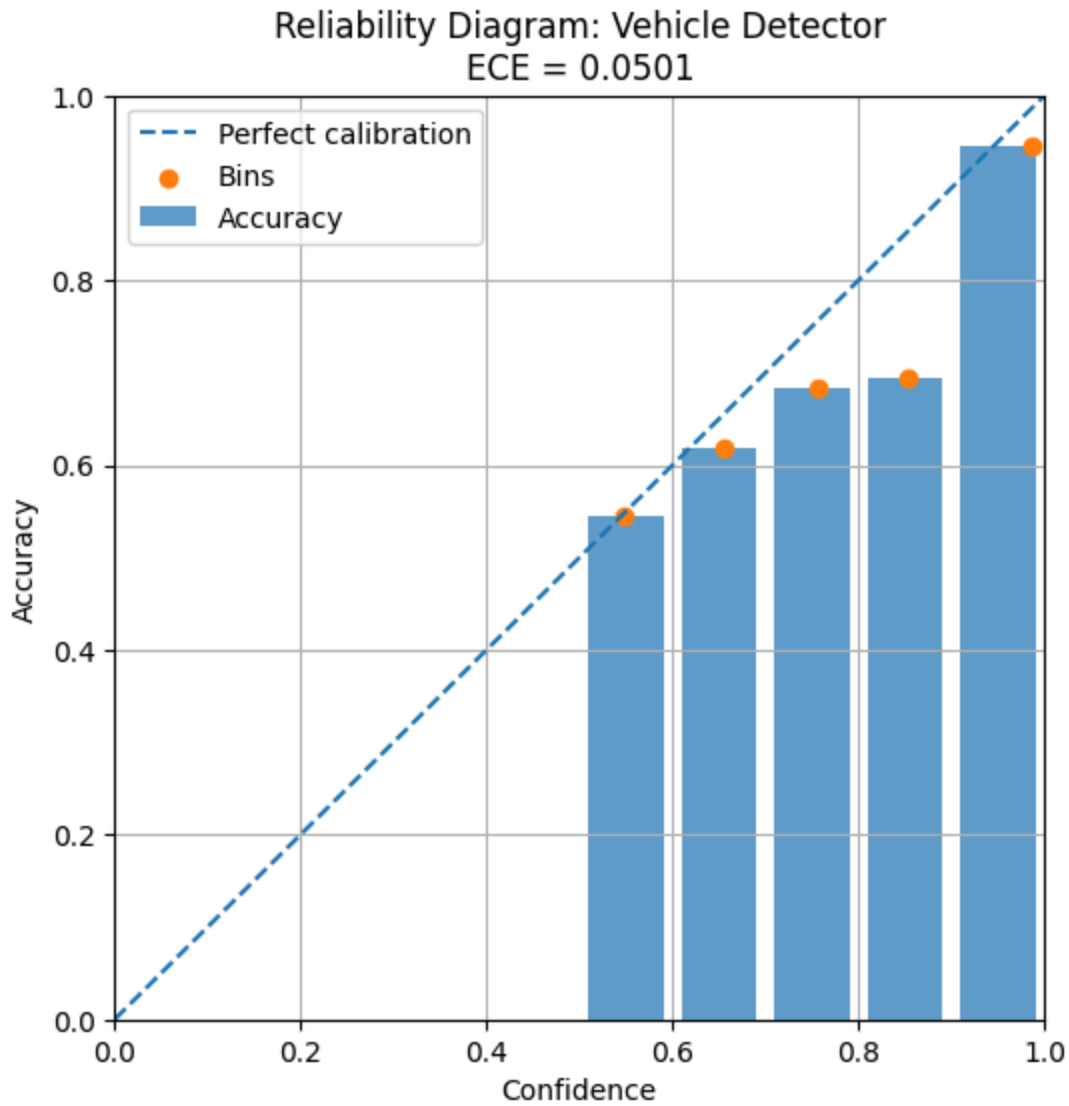
Reliability Diagram: Pedestrian Detector

ECE = 0.1322



Reliability Diagram: Traffic Light Detector
ECE = 0.0476





	Model	Accuracy	Mean Confidence	ECE
0	Pedestrian Detector	0.7900	0.9222	0.1322
1	Traffic Light Detector	0.9367	0.9811	0.0476
2	Vehicle Detector	0.8861	0.9363	0.0501

3. Are your models over- or underconfident? Does this pattern hold consistently across all three models?

→

Model	ECE	Calibration pattern	Interpretation
Traffic Light Detector	0.0476	Mostly calibrated	Best calibrated among the three
Vehicle Detector	0.0501	Slightly overconfident	Fairly well calibrated
Pedestrian Detector	0.1322	Overconfident	Worst calibrated

→ The Pedestrian Detector is the most miscalibrated and overconfident model. The Traffic Light and Vehicle detectors are better calibrated, although they still show some small calibration errors. Therefore, overconfidence is strongest for the pedestrian model and does not appear equally across all three models.

Overall, the calibration pattern is not fully consistent across all three models. The Traffic Light and Vehicle detectors are relatively well calibrated, while the Pedestrian Detector is noticeably overconfident and has the worst calibration.

Exercise 7.5: Temperature Scaling

Temperature scaling adjusts the model's logits as:
 $p(y | x) = \text{softmax}(f(x)/T)$

1. Apply temperature scaling to each of your three models. Optimize T on the validation set using negative log-likelihood.

→

NOTE: Temperature scaling is a calibration method. It does not retrain the model or change the predicted class. It only divides the model logits by a temperature value T before converting them into probabilities.

If $T > 1$, the probabilities become softer and less confident; if $T < 1$, they become sharper and more confident. We choose the best T on the validation set by minimizing negative log-likelihood.

NLL measures how much probability the model assigns to the true class and strongly punishes confident wrong predictions. After finding the best T, we compute ECE on the test set before and after scaling to check whether the model's confidence became better calibrated.

(ref ipynb for implementation)

2. Report the ECE before and after scaling for each model.

Hint: A simple line search is sufficient: evaluate the validation NLL for a grid of values such as $T \in \{0.5, 0.6, 0.7, \dots, 3.0\}$ (step 0.1) and pick the T with the lowest NLL.

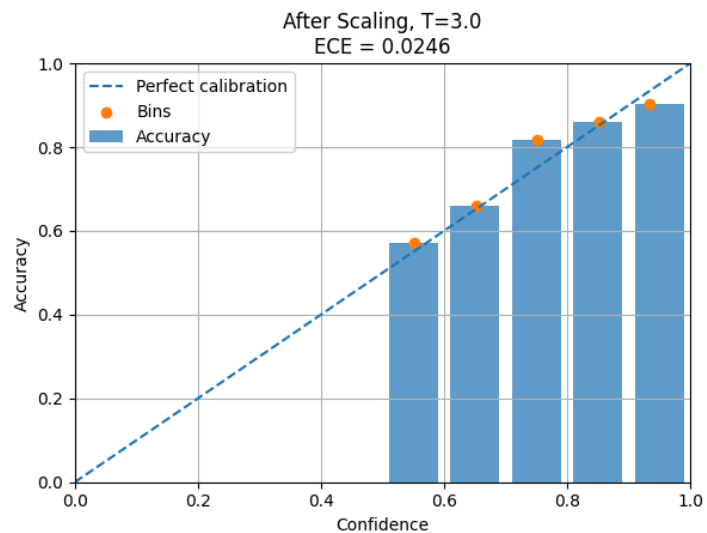
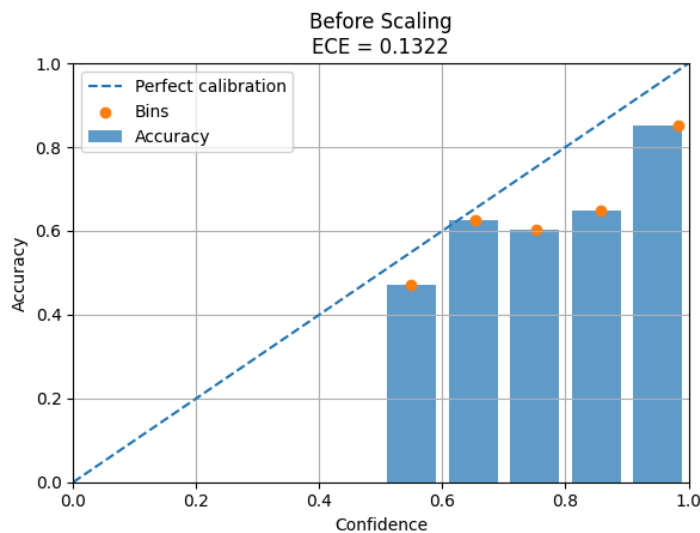
→

The Pedestrian Detector showed the largest improvement. Its ECE decreased from **0.1322** to **0.0246**, with the best temperature $T=3.0$. This indicates that the pedestrian model was strongly overconfident before calibration. The Traffic Light Detector improved slightly from **0.0476** to **0.0354**, with $T=1.4$. The Vehicle Detector improved from **0.0501** to **0.0135**, with $T=1.7$.

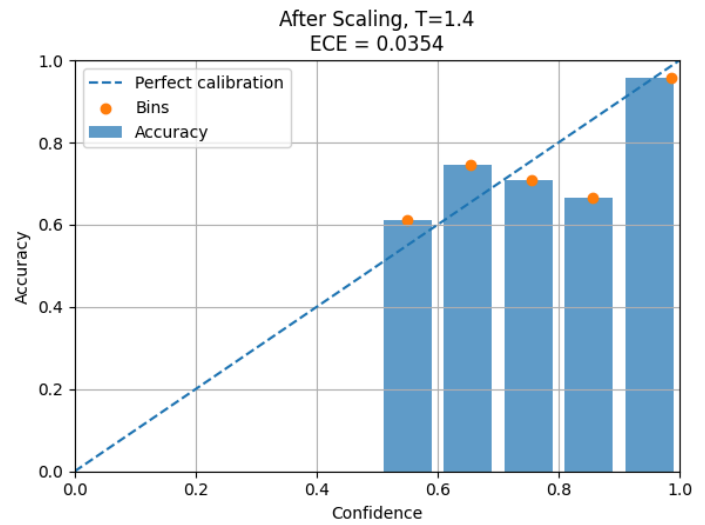
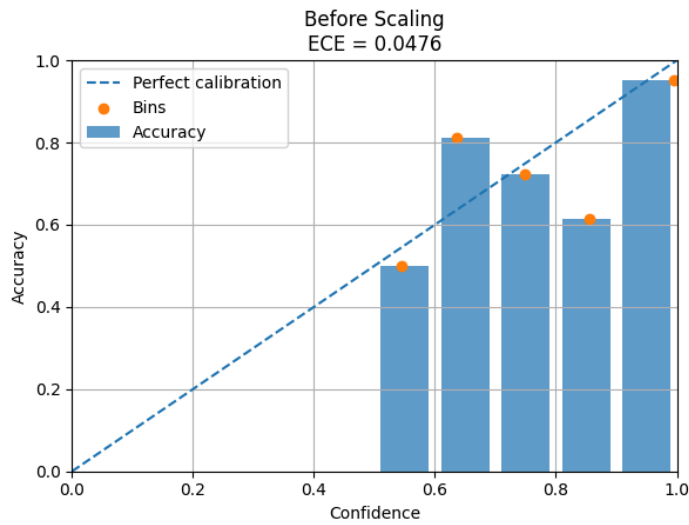
Overall, temperature scaling improved calibration for all three models. Since all best temperature values are greater than 1, all models were generally overconfident before scaling, especially the Pedestrian Detector.

Model	Best T	ECE before	ECE after	Improvement
Pedestrian Detector	3.0	0.1322	0.0246	0.1076
Traffic Light Detector	1.4	0.0476	0.0354	0.0122
Vehicle Detector	1.7	0.0501	0.0135	0.0366

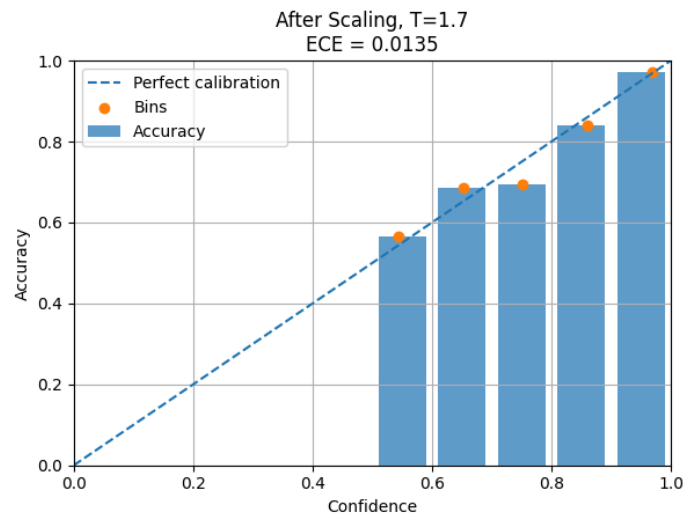
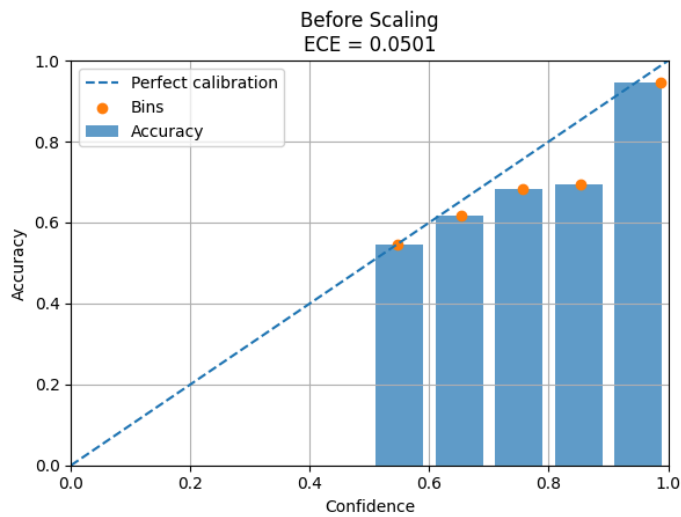
Reliability Diagram: Pedestrian Detector



Reliability Diagram: Traffic Light Detector



Reliability Diagram: Vehicle Detector



Exercise 7.6: Cost-Optimal Decision in Practice

Use the costs from Exercise 3 (CFN = 100, CFP = 1, $\tau^* \approx 0.0099$, i.e. a predicted probability of roughly 1 %) and apply them to your pedestrian classifier.

1. On the in-distribution test set, compute the total loss

$$L = \text{CFN} \cdot \text{\#FN} + \text{CFP} \cdot \text{\#FP}$$

for your uncalibrated model at both $\tau = 0.5$ and $\tau = \tau^*$

→ (refer ipynb for implementation)

2. Repeat for your temperature-scaled model at both thresholds.

→ (refer ipynb for implementation)

3. Arrange the four results in a 2×2 table (rows: uncalibrated / calibrated; columns: $\tau = 0.5$ / $\tau = \tau^*$). Which combination gives the lowest total loss?

→

Model	$\tau^* = 0.0099$	$\tau = 0.5$
Calibrated	2993	46989
Uncalibrated	19110	46989

→ The lowest total loss was obtained by the calibrated model with τ^* , giving a loss of 2993. The uncalibrated model with τ^* also improved over the standard threshold, but its loss was still much higher at 19110. Both calibrated and uncalibrated models had a high loss of 46989 at $\tau=0.5$, showing that the default threshold is not suitable for this safety-critical task.

Overall, the best decision rule is to use the temperature-scaled pedestrian model with the cost-optimal threshold.

Exercise 7.7: Tracing Overconfidence Through the Safety Analysis

You have now measured how miscalibrated confidence propagates silently to downstream decisions. In STPA, an overconfident misclassification is a causal factor: a mechanism that explains why an existing unsafe control action occurs. The goal of this exercise is to add it as a loss scenario and trace it through to a verifiable constraint.

1. Causal scenario. Revisit your causal loss scenarios from Exercise 2.8. The hazard (“vehicle fails to brake for a pedestrian”, Exercise 2.4) and the unsafe control action (“the planner does not command braking while a pedestrian is in the path”, Exercise 2.6) already exist - you do not need new ones. Add a causal scenario describing a new way that UCA arises: the pedestrian classifier produces a false negative yet reports high confidence, so the planner registers “no pedestrian, reliable” and no low-confidence fallback is triggered.

→

UCA	Causal Factor	Why It Occurs / Root Cause	Related Safety Constraint
UCA-13: Planning module does not issue a brake command while a pedestrian is in the vehicle path	Pedestrian classifier produces a false negative but reports high confidence	The pedestrian is present, but the perception model predicts “no pedestrian” with high confidence due to miscalibration. Because the prediction appears reliable, the planner registers the scene as “no pedestrian, reliable” and no low-confidence fallback is triggered	SC-13: The system shall not rely only on raw classifier confidence for pedestrian absence. If pedestrian probability, uncertainty, or calibration-aware confidence indicates possible risk, the planner shall trigger braking, slow down, or activate a safe fallback

2. Safety constraint. From the scenario, derive at least one new safety constraint of each kind (cf. Exercise 2.7):

- a model-level constraint on calibration, expressed with the ECE you measured; and
- a system-level constraint on how the planner uses the confidence score - e.g. the fallback when confidence is low, or adopting the cost-optimal threshold τ^* from Exercise 3 instead of 0.5.

→

Constraint ID	Type	Safety constraint
SC-13	Model-level calibration constraint	The pedestrian classifier shall be calibrated before deployment. On the in-distribution validation/test set, its ECE shall remain below a defined safety limit, for example $ECE \leq 0.05$. The uncalibrated pedestrian model had $ECE = 0.1322$, which violates this constraint, while the temperature-scaled model reduced ECE to 0.0246, satisfying the constraint.

Constraint ID	Type	Safety constraint
SC-14	System-level planner constraint	The planner shall not use the default threshold $T = 0.5$ for pedestrian braking decisions. Since false negatives are safety-critical and ($C_{\{FN\}}=100$), the planner shall use the cost-optimal threshold $T^* = 0.0099$, or an equivalent conservative fallback rule. If the calibrated pedestrian probability exceeds T^* , the planner shall brake, slow down, or trigger a safe fallback.

3. Verification. What evidence verifies the model-level constraint? You produced exactly this evidence on this sheet (the ECE before and after temperature scaling, Exercise 7.4–Exercise 7.5). Does your calibrated model meet the threshold you set?

→

The evidence verifying the model-level constraint is the ECE evaluation performed in Exercises 7.4 and 7.5. In Exercise 7.4, the uncalibrated pedestrian classifier had an ECE of **0.1322**, showing that its confidence values were not reliable enough for safety-critical use. After applying temperature scaling in Exercise 7.5, the pedestrian classifier's ECE decreased to **0.0246**.

Since the safety constraint requires **$ECE \leq 0.05$** , the uncalibrated model does not satisfy the constraint, but the calibrated model does. Therefore, the temperature-scaled pedestrian classifier meets the model-level calibration requirement.

4. Residual risk. Even with a perfectly calibrated model, what does the constraint not cover? Does calibration alone close the scenario, or is the system-level fallback still required?

→

Even with a perfectly calibrated model, the constraint does not guarantee that the pedestrian classifier will always be correct. Calibration only ensures that the reported confidence matches empirical accuracy on average.

It does not eliminate false negatives, rare corner cases, occlusion, poor lighting, sensor faults, distribution shift, or delayed detection.

Also, ECE is an aggregate metric, so it does not guarantee safety for every individual frame.

It reduces the risk of silent overconfident misclassification, but a PROPER system-level fallback is still required and its working should be ensured.